

RATE & EXTENT – REVERSIBLE REACTIONS

KEY VOCABULARY

Reversible reaction: products can react to reform the reactants ().

Forward reaction: reactants → products.

Backward reaction: products → reactants.

Closed system: nothing enters or leaves the reaction.

COMMON MISCONCEPTIONS

~~A reversible reaction goes one way then stops.~~

✓ Both directions happen at the same time.

~~If forward is exothermic, so is backward.~~

✓ The backward reaction is endothermic by the same amount.

~~Reversible reactions reach 100% yield.~~

✓ They settle at an equilibrium below 100%.

ENERGY IN BOTH DIRECTIONS

Hydrated copper sulfate anhydrous + water.

Forward (heating) is endothermic.

Backward (adding water) is exothermic.

Energy in = energy out (same magnitude).

The symbol shows a reversible reaction.

1

THE SYMBOL

State what the symbol means in an equation. [1 mark]

2

ENERGY

The forward reaction is exothermic. **State** the energy change of the backward reaction and why. [2 marks]

3

COPPER SULFATE

Describe the colour change when hydrated copper sulfate is heated. [2 marks]

4

REACH EQUILIBRIUM

Explain why a reversible reaction does not give a 100% yield. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

If the forward reaction is _____, the backward reaction is _____.

endothermic

exothermic

forward

backward

6

APPLY

Suggest how copper sulfate could be used to test for water. [2 marks]